

SUPPLEMENT TO “IDENTIFYING RATIONAL TYPES IN UNKNOWN
ENVIRONMENTS”
APPENDIX 4: STREAMLIT CALIBRATION MODEL

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S1. SCOPE OF THIS COMPUTATIONAL APPENDIX

This appendix documents only the equations that are used by the Streamlit calibration app in `./Streamlit/app.py`. It follows the order of the main manuscript: first the empirical objects that discipline priors and hazards, then the theoretical mechanism implemented in the app, then the feasibility diagnostics, and finally the way in which these equations generate Table 5.2.

The macroeconomic VEC and LLT estimates in the main paper are not reproduced here because the Streamlit app does not re-estimate them and does not use them directly to compute Table 5.2. The equations below are the active estimated or calibrated objects used by the app.

S2. EMPIRICAL INPUTS USED BY THE APP

S2.1. Outcome Model and Prior Over Types

The first empirical input is the multinomial outcome model from the manuscript. With payment as the reference outcome, the estimated equation is

$$\ln\left(\frac{\Pr(Y_{i,m} = j \mid \mathbf{X}_{i,m})}{\Pr(Y_{i,m} = 1 \mid \mathbf{X}_{i,m})}\right) = \alpha_j + \mathbf{G}'_i \boldsymbol{\beta}_{1,j} + \mathbf{V}'_i \boldsymbol{\beta}_{2,j} + \mathbf{C}'_{i,m} \boldsymbol{\beta}_{3,j} + \mathbf{T}'_i \boldsymbol{\beta}_{4,j}, \quad j \in \{0, 2, 3\}. \quad (\text{S2.1})$$

In the app, this block is used to discipline the initial type probabilities and case covariate adjustments. The prior used by the Streamlit state is a softmax:

$$\mu_0(\theta \mid z, \theta_V) = \frac{\exp(\varpi_\theta + \eta_{\theta,z} + \xi_{\theta,\theta_V})}{\sum_{\theta' \in \Theta_K} \exp(\varpi_{\theta'} + \eta_{\theta',z} + \xi_{\theta',\theta_V})}. \quad (\text{S2.2})$$

The inputs are the type baseline ϖ_θ , the region adjustment $\eta_{\theta,z}$, and the victim-profile adjustment ξ_{θ,θ_V} . The output is the initial posterior vector μ_0 used by Table 5.2.

S2.2. Cause-Specific Duration Model

The second empirical input is the Cox competing-risk block. For each cause j , the manuscript estimates

$$h_j(t \mid \mathbf{X}) = h_{0j}(t) \exp(\mathbf{X}' \tilde{\boldsymbol{\beta}}_j), \quad S_j(t \mid \mathbf{X}) = \exp\left[-H_{0j}(t) \exp(\mathbf{X}' \tilde{\boldsymbol{\beta}}_j)\right]. \quad (\text{S2.3})$$

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The app uses this information through baseline risks λ_{j0} and type-cause coefficients $\beta_{K,j}(\theta)$, stored in the model object and updated by the calibration interface. These coefficients feed the structural intensities in Section S3.3.

S3. THEORETICAL MODEL IMPLEMENTED IN STREAMLIT

S3.1. State, History, and Cycle

The app implements the paper’s dynamic history as the period-by-period record of instruments, executed actions, physical outcomes, and detection signals:

$$h_t = (t, \alpha_{[0,t-1]}^*, \gamma_{[0,t-1]}^*, \tilde{a}_{[0,t-1]}^F, \tilde{a}_{[0,t-1]}^K, \tilde{a}_{[0,t-1]}^S, m_{[0,t-1]}, d_{[0,t-1]}) , \quad (\text{S3.1})$$

$$h_{t+1} = (h_t, \alpha_t^*, \gamma_t^*, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, m_t, d_t) . \quad (\text{S3.2})$$

The terminal event is

$$\tau = \inf\{t \geq 0 : m_t \in \mathcal{M}_{\text{term}}\} . \quad (\text{S3.3})$$

When Table 5.2 runs in realized-draw mode, the simulated event stops after the first $m_t \neq \text{cont}$; the diagnostic tables still retain the probability vector used to generate the draw.

S3.2. MDG Implementation Kernel

For each agent $i \in \{K, S, F\}$, the app maps the intended action a_t^{i*} into an executed action $\tilde{a}_t^i$ with the MDG logit kernel:

$$\mathbb{P}_{I,i}(\tilde{a}_t^i = a \mid a_t^{i*}, X_t) = \frac{\exp\{\mathbf{1}(a = a_t^{i*})/T_t\}}{\sum_{a' \in \mathcal{A}^i} \exp\{\mathbf{1}(a' = a_t^{i*})/T_t\}} . \quad (\text{S3.4})$$

The temperature used in the app is

$$T_t = T_0 \max\left\{\frac{H(\mu_t)}{H(\mu_0)} e^{-\eta_{\text{cal}} t}, \bar{c}\right\}, \quad H(\mu_t) = - \sum_{\theta \in \Theta_K} \mu_t(\theta) \ln \mu_t(\theta). \quad (\text{S3.5})$$

The app also reports the bounded operational noise

$$\varepsilon_t = \min\{0.5, \max\{0, \varepsilon_0 T_t\}\} . \quad (\text{S3.6})$$

For Table 5.2, the implemented-action likelihood is

$$\mathcal{L}_t^I(\theta) = \prod_{i \in \{K, S, F\}} \mathbb{P}_{I,i}(\tilde{a}_t^i \mid a_t^{i*}, X_t) , \quad (\text{S3.7})$$

computed by the implementation-likelihood block in `app.py`.

S3.3. Competing-Risk Technology

The app evaluates the structural proportional-risk equation

$$\lambda_j(t \mid \mathcal{C}_t, \theta) = \lambda_{j0}(t) \exp(x_t' \beta_j(\theta)), \quad j \in \{1, 2, 3\}. \quad (\text{S3.8})$$

Operationally, the app uses $\tilde{\lambda}_j(t | \theta_K) = M(t)\lambda_{j0} \exp(E_j)$, with

$$E_1 = \beta_{K,1}(\theta_K) + \beta_{z,1} + \beta_{F,1} - \beta_{V,1} + \beta_{S,1} - \zeta_{\alpha,1}\alpha_t - \zeta_{\gamma,1}\gamma_t - \zeta_{d,1}p_{\text{det},t} + \phi_{F,1}\mathbf{1}_F + \phi_{K,1}^{\text{cont}}\mathbf{1}_K^{\text{cont}}, \quad (\text{S3.9})$$

$$E_2 = \beta_{K,2}(\theta_K) + \beta_{z,2} + \beta_{S,2} + \zeta_{\alpha,2}\alpha_t + \zeta_{\gamma,2}\gamma_t - \zeta_{d,2}p_{\text{det},t} - \phi_{F,2}\mathbf{1}_F + \phi_{K,2}^{\text{kill}}\mathbf{1}_K^{\text{kill}} + \phi_{K,2}^{\text{cont}}\mathbf{1}_K^{\text{cont}}, \quad (\text{S3.10})$$

$$E_3 = \beta_{K,3}(\theta_K) + \beta_{z,3} - \beta_{S,3} + \zeta_{\alpha,3}\alpha_t + \zeta_{\gamma,3}\gamma_t + \zeta_{d,3}p_{\text{det},t} + \zeta_R\mathbf{1}_S^{\text{res}} - \phi_{F,3}\mathbf{1}_F + \phi_{K,3}^{\text{cont}}\mathbf{1}_K^{\text{cont}}. \quad (\text{S3.11})$$

Equivalently,

$$\tilde{\lambda}_j(t | \theta_K) = M(t)\lambda_{j0} \exp(E_j), \quad j = 1, 2, 3. \quad (\text{S3.12})$$

Here

$$M(t) = \min\{1, (t/T_{\text{mad}})^2\}, \quad \tilde{\lambda}_4(t) = \lambda_4. \quad (\text{S3.13})$$

The daily probabilities used by Table 5.2 are computed as

$$p_{\text{Cont},t|\theta_K} = \exp\left[-\Delta t \sum_{j=1}^4 \tilde{\lambda}_j(t | \theta_K)\right], \quad (\text{S3.14})$$

$$q_t(\theta_K) = 1 - p_{\text{Cont},t|\theta_K}, \quad (\text{S3.15})$$

$$\xi_j(t | \theta_K) = \frac{\tilde{\lambda}_j(t | \theta_K)}{\sum_{\ell=1}^4 \tilde{\lambda}_\ell(t | \theta_K)}, \quad h_j(t | \theta_K) = q_t(\theta_K)\xi_j(t | \theta_K). \quad (\text{S3.16})$$

These calculations are performed by `mechanism_competitive_hazards_at_t`. The returned vector is

$$(p_{\text{Cont}}, h_{\text{pay}}, h_{\text{kill}}, h_{\text{res}}, h_{\text{rel}}).$$

S3.4. Detection and Communication Signals

The detection probability used by the app is type-specific in its intercept:

$$p_{\text{det},t}(\theta_K) = \Lambda(\eta_0(\theta_K) + \eta_1\alpha_t^* + \eta_2\gamma_t^*), \quad \Lambda(u) = \frac{1}{1 + \exp(-u)}. \quad (\text{S3.17})$$

The detection likelihood is Bernoulli:

$$\mathcal{L}_t^d(d_t | \theta_K) = p_{\text{det},t}(\theta_K)^{d_t} [1 - p_{\text{det},t}(\theta_K)]^{1-d_t}. \quad (\text{S3.18})$$

When the communication block is active, the app evaluates either a voice likelihood or a silence likelihood. The voice signal is

$$x_t^{\text{obs}} = x_t^{\text{true}}(\theta_K) + \varepsilon_L + \varepsilon_S, \quad \varepsilon_L \sim N(0, \Sigma_L), \quad \varepsilon_S \sim N(0, \Sigma_S), \quad (\text{S3.19})$$

with diagonal likelihood

$$\mathcal{L}_{\text{voz},t}(\theta_K) \propto \exp\left[-\frac{1}{2} \sum_k \left(\frac{x_{t,k}^{\text{obs}} - m_k(\theta_K)}{\sigma_k(\theta_K)}\right)^2\right]. \quad (\text{S3.20})$$

If no call is observed, the app uses

$$\mathcal{L}_t^C(\theta_K \mid V_t = 0) = [1 - \pi_{\text{call}}(\theta_K)]^{\omega_{\text{voz}}}. \quad (\text{S3.21})$$

S3.5. Physical Likelihood and Posterior

Given the competing-risk vector, the physical likelihood is

$$\mathcal{L}_t^H(m_t \mid \theta_K) = p_{\text{Cont},t|\theta_K}^{\mathbf{1}\{m_t=\text{cont}\}} \prod_{j=1}^4 h_j(t \mid \theta_K)^{\mathbf{1}\{m_t=g(j)\}}, \quad (\text{S3.22})$$

where $g(1) = \text{pay}$, $g(2) = \text{kill}$, $g(3) = \text{res}$, and $g(4) = \text{rel}$. The total likelihood used for Table 5.2 is

$$\mathcal{L}_t(\theta_K) = \mathcal{L}_t^I(\theta_K) \mathcal{L}_t^H(m_t \mid \theta_K) \mathcal{L}_t^d(d_t \mid \theta_K) \mathcal{L}_t^C(\theta_K). \quad (\text{S3.23})$$

The posterior update is

$$\mu_{t+1}(\theta_K) = \frac{\mu_t(\theta_K) \mathcal{L}_t(\theta_K)}{\sum_{\theta' \in \Theta_K} \mu_t(\theta') \mathcal{L}_t(\theta')}. \quad (\text{S3.24})$$

This is computed by `build_t0_bayesian_posterior_report` and `bayesian_posterior_update`. The output is the likelihood decomposition and the new belief vector used in the next cycle.

S4. POLICY AND FEASIBILITY EQUATIONS USED BY THE APP

S4.1. State Objective

The app computes the state's branch loss as

$$L_t(a_t^S, \alpha_t, \gamma_t, \theta_K, \iota_t) = \begin{cases} V_t^R(\iota_t, \hat{\theta}_t, \theta_K, \alpha_t, \gamma_t), & a_t^S = a_{\text{res}}, \\ V_t^N(\theta_K, \alpha_t, \gamma_t), & a_t^S = a_{\text{neg}}. \end{cases} \quad (\text{S4.1})$$

The numerical optimizer for Table 5.2 solves

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{\Gamma_t(\mu_t)} \left[\sum_{\theta \in \Theta_K} \mu_t(\theta) L_t(a_t^S, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi_{t,a^S}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \right]. \quad (\text{S4.2})$$

The penalty around the Bayesian benchmark and the entropy gain are

$$\Pi_{t,b}^S(\alpha_t, \gamma_t; \mu_t) = \chi_\alpha(\alpha_t - \alpha_{t,b}^\mu)^2 + \chi_\gamma(\gamma_t - \gamma_{t,b}^\mu)^2, \quad (\text{S4.3})$$

$$\Delta H_t(\alpha_t, \gamma_t) = H(\mu_t) - \sum_m \mathbb{P}(m_t = m \mid \mu_t, \alpha_t, \gamma_t) H(\mu_{t+1}^m). \quad (\text{S4.4})$$

The app compares the rescue and negotiation floors and selects rescue when $\tilde{V}_t^{R,*} \leq \tilde{V}_t^{N,*}$.

TABLE I
EQUATION-TO-OUTPUT MAP FOR TABLE 5.2

Table 5.2 object	Equation used by the app
Prior and posterior beliefs	Prior softmax (S2.2) and Bayes rule (S3.24).
Optimal state instruments	State objective (S4.2), penalty (S4.3), and entropy gain (S4.4).
MDG intended/executed actions	Kernel (S3.4) and implementation likelihood (S3.7).
Physical outcome probabilities	Intensities (S3.12) and probabilities (S3.14)–(S3.16).
Detection signal	Logistic detection (S3.17) and Bernoulli likelihood (S3.18).
Posterior decomposition	Physical likelihood (S3.22), total likelihood (S3.23), and normalization in (S3.24).
Feasibility diagnostics	IC^K , IR^K , and IR^F in (S4.8)–(S4.10).

S4.2. Private-Agent Equations and Diagnostics

The app evaluates the captor’s continuation cost as

$$C_t(\gamma_t, \theta_K) = \phi(\theta_K) \exp\{\kappa_c(\theta_K)\gamma_t\} + \nu(\theta_K). \quad (\text{S4.5})$$

The family compares cooperation with clandestine payment:

$$\mathcal{U}_t^F(a_{\text{coop}}) = \mathbb{E}_{\theta|\mathcal{I}_t^F} [\tilde{p}_{\text{surv},t}(\theta)] V_L - e_t(\gamma_t, \theta_F), \quad (\text{S4.6})$$

$$\mathcal{U}_t^F(a_{\text{col}}) = \mathbb{E}_{\theta|\mathcal{I}_t^F} [\tilde{p}_{\text{rel},t}(\theta)] V_L - R - p_{\text{det},t} F_{\text{col}}. \quad (\text{S4.7})$$

The feasibility diagnostics shown with Table 5.2 and Tab 6 are

$$IC_t^K : \quad \mathbb{E}_{\theta \sim \mu_t} [V_t^K(a^*(\theta) \mid \theta, \alpha_t, \gamma_t) - V_t^K(a^*(\theta_j) \mid \theta, \alpha_t, \gamma_t)] \geq 0, \quad (\text{S4.8})$$

$$IR_t^K : \quad \mathbb{E}_{\theta \sim \mu_t} [U_{\text{rel}}^K(\theta) - \max\{V_{\text{cont},t}^K(\theta, \alpha_t, \gamma_t), U_{\text{kill}}^K(\theta, \theta_S)\}] \geq 0, \quad (\text{S4.9})$$

$$IR_t^F : \quad \mathcal{U}_t^F(a_{\text{coop}}) \geq \mathcal{U}_t^F(a_{\text{col}}). \quad (\text{S4.10})$$

S5. HOW THE APP OBTAINS TABLE 5.2

Table 5.2 is produced by iterating the equations above. The sequence implemented in `app.py` is:

$$\mu_t \rightarrow (a_t^{K*}, a_t^{S*}, a_t^{F*}) \rightarrow \tilde{A}_t \rightarrow (\tilde{\lambda}_j, p_{\text{Cont}}, h_j, p_{\text{det}}) \rightarrow (m_t, d_t, V_t, x_t^{\text{obs}}) \rightarrow \mu_{t+1}. \quad (\text{S5.1})$$

At $\tau = 0$, the app starts from μ_0 in (S2.2). It computes the state policy $(a_0^{S*}, \alpha_0^*, \gamma_0^*)$ from (S4.2), computes the private intended actions, applies the MDG implementation kernel (S3.4), evaluates p_{det} using (S3.17), computes the physical probabilities through (S3.14)–(S3.16), and then forms the likelihood (S3.23). The posterior column μ_1 is obtained from (S3.24).

For each subsequent cycle $\tau \geq 1$, the posterior inherited from the previous cycle becomes the new prior. The same optimizer, implementation kernel, hazard block, detection block, likelihood block, and Bayes rule are applied again. Thus, every column of Table 5.2 is a repeated application of (S5.1).

The resulting objects are stored by the app in the dynamic Table 5.2 session state, including `dynamic_cycles52`, `dynamic_cycles_diag52`, `dynamic_cycles_stop52`, `first_cycle_table52`, and `first_cycle_post54`. Tab 6 reads these same objects; it does not change the calculations used to produce Table 5.2.